

PostgreSQL

SQL QUERIES FOR DATA EXTRACTION

```
/* SOXX*/  
SELECT  
tickerdate,  
tickertime,  
open,  
high,  
low,  
close,  
vw_price,  
volume,  
trades  
FROM polygon_1m_bars.soxx  
WHERE tickerdate BETWEEN '2021-01-01' AND '2023-12-31'  
AND tickertime BETWEEN '09:30:00' AND '16:00:00'  
ORDER BY tickerdate, tickertime;
```

R STUDIO : Convert date & time into proper formats

```
library(dplyr)
library(lubridate)
```

```
fix_data <- function(df) {
  df %>%
  mutate(
    tickerdate = as.Date(tickerdate),
    tickertime = hms::as_hms(tickertime),
    hour = hour(tickertime),    # Extract hour of day
    minute = minute(tickertime),
    price_change = close - open, # Short-term movement
    volatility = high - low     # Intrabar range
  )
}
```

```
nvda <- fix_data(nvda)
avgo <- fix_data(avgo)
```

R STUDIO Aggregate by time of day across 3 years

```
nvda_summary <- nvda %>%  
  group_by(time_of_day = format(time_bin, "%H:%M")) %>%  
  summarise(  
    avg_volume = mean(volume, na.rm = TRUE),  
    avg_volatility = mean(volatility, na.rm = TRUE)  
  )
```

```
avgo_summary <- avgo %>%  
  group_by(time_of_day = format(time_bin, "%H:%M")) %>%  
  summarise(  
    avg_volume = mean(volume, na.rm = TRUE),  
    avg_volatility = mean(volatility, na.rm = TRUE)  
  )
```

R STUDIO : Create heatmaps in RStudio (ggplot2)

```
library(ggplot2)
```

```
# NVDA Volume Heatmap
```

```
ggplot(nvda_summary, aes(x = time_of_day, y = "NVDA", fill = avg_volume)) +  
  geom_tile() +  
  scale_fill_viridis_c() +  
  labs(title = "NVDA — Intraday Volume Heatmap",  
        x = "Time of Day", y = "") +  
  theme(axis.text.x = element_text(angle=90, hjust=1))
```